A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 36

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Tropical intertidal coastal forests of halophytic (salt-tolerant) trees in saline, oxygen-deficient (anaerobic) mud **1** Prop / aerial roots for support in soft sediment and for gas exchange (pneumatophores) – salt excretion **1** Seeds germinate on the parent tree and float, rooting where they touch sand - rapid natural regeneration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 4 points below. **1 Coastal erosion / storm protection:** roots and vegetation absorb wave energy (least tsunami damage in 2004 where mangroves were intact) **1 Fisheries:** nursery grounds where dense roots protect young fish that later live on reefs / open sea **1 Protection of coral reefs:** slow river water so suspended solids settle and nutrients are absorbed before reaching reefs **1 Resources:** timber, fuel – medicines (antimicrobials against MRSA, cyanobacterium for lung cancer) – high biodiversity (crabs, shrimps, sponges, birds) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
01.3	**1 mark each** 1 mark for each of the 3 points below. **1** Clearance for aquaculture (shrimp ponds), urban development, ports, tourism – pollution – sea level rise / climate change – destruction of coral reefs that shelter the coast **1** Legal protection / designated areas (eg Sri Lanka protected all remaining mangroves in 2015) **1** Replanting and natural regeneration – control of damaging activities – Ramsar sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 66-67; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 4 points below. **1** Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis – human processes usually use high energy-density non-renewable fossil fuels **1** Natural processes occur at low temperatures because enzymes lower the activation energy of reactions – human processes need high temperatures and pressures **1** Natural processes produce no toxic wastes / their products are raw materials for other processes – industrial processes emit CO2 and NOx **1** So natural fixation is sustainable indefinitely with a low carbon footprint **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4
02.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Hydrological cycle (evaporation, precipitation) **1** Carbon cycle / photosynthesis **1** Atmospheric circulation / winds **1** Thermohaline circulation (density differences from heating / evaporation) **1** Decomposition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration	3
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting **1** Anaerobic digestion and composting treat wastes – enzymes in industry (biological washing powders, biofuels from algae / microorganisms) **1** Heat pumps and solar thermal capture low-density heat – low-temperature manufacturing using biological polymers – fuel cells instead of combustion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Coordinating global data on biodiversity conservation (Red List) **1** Increasing understanding of the importance of biodiversity **1** Deploying nature-based solutions to global challenges in climate, food and sustainable development **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Decline = (12 000 - 2100) / 12 000 x 100 = 82.5% **1** Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km^2) and adults (2100) only meet Endangered **1** A species is placed in the highest category for which it meets any one criterion, so Critically Endangered **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.3	**1 mark each** 1 mark for each of the 3 points below. **1** Categorising a species requires a great deal of research (population, range, trends) and resources – about 85 000 of millions of species assessed **1** Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests) **1** Evolutionarily unique species / EDGE species – endemic species – keystone species – flagship species – species with fragmented populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **endemic** → found naturally only in that one place • **keystone species** → a species whose loss changes the whole community	3

Do not write outside the box

Question	Marking guidance	Marks
03.4	1 mark for the point below. **1 Iberian lynx:** Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years) – Fregate Island beetle after rat eradication **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.	
04.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 1000 / 100 000 x 100 = 1% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Some light is reflected by the leaves **1** Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed **1** Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) **1** After four levels too little energy remains to support a further population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) **1** Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **symbiotic** → a close relationship in which both species benefit **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
04.5	**1 mark each** 1 mark for each of the 2 points below. **1** Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) **1** So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
05.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 2.7 - 2.45 = 0.25 billion = 250 million years **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (million years) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans **1** Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Oxygen in the atmosphere absorbed ultraviolet light **1** A dynamic equilibrium of reactions between O, O_2 and O_3 (the ozone layer) developed **1** Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) **1** Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved – many anaerobic Archaea and bacteria died out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) **1** The lower CO_2 concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion **1** Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas **1** Regulation of atmospheric composition (eg oxygen / CO_2 by photosynthesis and respiration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Transpiration** → water lost as vapour from the leaves **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 40-50 traps **1** Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Widely scattered results (large standard deviation) mean a single sample may not be representative **1** More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
06.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Same type and size of pitfall trap, set for the same length of time **1** Traps emptied at the same time of day / same weather conditions in both habitats **1** Same number of traps and same method of positioning (random or systematic) in each habitat **1** Other variables that could affect results (eg rainfall, temperature) monitored or controlled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **pitfall trap** → a sunken container that catches ground invertebrates • **random** → positions chosen using random numbers so the choice is not biased	3
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally **1** Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends **1** The interval between samples should match the rate at which the factor changes, found from a preliminary study **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable **1** If it cannot be adjusted the percentage error should be recorded so the results can be corrected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	2
07.2	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1 Droppings:** diet, gender, territory size **1 Owl pellets:** diet **1** Feeding marks (eg chewed hazel nuts): presence of species such as dormice **1 Nests / burrows:** presence and breeding **1 Tracks / footprints:** presence, movements **1** Territorial marks (scratching posts): presence and territory **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species **1** The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Marked individuals mix freely with the rest of the population **1** Marks do not fall off or wear away, and marking does not affect survival **1** The population does not change during the study (no births, deaths or migration) **1** All individuals have an equal chance of being caught / animals are mobile and can be caught **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mark-release-recapture** → estimating population size from the proportion of marked animals recaptured **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
08.1	**1 mark each** 1 mark for each of the 4 points below. **1** Very small gene pool so individuals are closely related and inbreeding is likely **1** Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low **1** A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) **1** Exchange of individuals / semen between zoos – avoid hybridisation with related species / subspecies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
08.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided **1 Population interactions:** mixing breeding and non-breeding individuals causes conflict – some need a choice of partners, others pair for life **1 Breeding habitat needed:** courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) **1** Stress / disturbance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
08.3	**1 mark each** 1 mark for each of the 3 points below. **1 AI:** semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet – avoids dangerous mating (elephants, rhinos) and partner rejection **1 Embryo transfer:** a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) **1** So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Hedgerows **1** Hay meadows **1** Grazed moorland / heathland **1** Chalk grassland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive **1** Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
09.3	1 mark for the point below. **1** The introduced (non-indigenous) predator may eat other species as well as the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Context:** wild catches have not increased since about 1990 as stocks are fully or over-exploited – aquaculture output growing rapidly and approaching the wild catch – fish are the last large-scale hunted food • **Advantages of aquaculture:** control of limiting factors, selective breeding, gender control, high yields per area – fish have low FCRs (no thermoregulation / support) compared with livestock – herbivores (tilapia, carp) and molluscs very efficient – polyculture, IMTA, aquaponics, rice-fish • **Limits:** carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins) – some species (bluefin tuna) will not breed in captivity – consumer preference for carnivores • **Environmental impacts:** habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control – and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits – MSY management needs good data • **Judgement:** aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing – both need sustainable management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquaculture** → fish or shellfish farming	9
	• **gene pool** → the range of alleles / genes present in a population • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Wild catches have not increased since about 1990 because most stocks are fully or over-exploited, while aquaculture has grown rapidly and now approaches the wild catch, so the question is how far it can take over. Its advantages are real: limiting factors, breeding, gender and disease can be controlled, yields per area are high, and fish have low food conversion ratios because they need not support their weight or keep warm; herbivores such as tilapia and carp and filter-feeding molluscs need no fishmeal at all.	
	Its limits are equally real. The carnivorous species consumers prefer - salmon, shrimp - are fed fishmeal from wild forage fish, so their farming still depends on fishing and depletes species such as sandeels on which puffins rely; some species, such as bluefin tuna, will not breed in captivity. Farms destroy mangroves, release organic waste, pesticides and antibiotics that breed resistance, spread lice, allow escapes that contaminate wild gene pools and provoke predator control, though siting, closed systems, polyculture and vaccines can reduce these. Meanwhile fishing itself can be sustainable if quotas based on good stock data, gear design, size and effort limits and no-take zones are enforced.	
	Aquaculture can therefore supply most of the growth in demand, especially through herbivorous fish and shellfish, and can relieve pressure on wild stocks, but it cannot fully replace fishing while carnivorous species depend on wild fishmeal and some species cannot be farmed. Both must be managed sustainably; neither is a substitute for the other.	
	Weaker answer - Level 1 (2 marks)	
	Aquaculture is farming fish instead of catching them. Fish farms can produce a lot of fish and they do not run out like wild fish do. But fish farms cause pollution and diseases can spread to wild fish. Salmon farms feed the salmon on fish which have to be caught. So aquaculture can replace some fishing but not all of it.	
	What makes the difference	
	• Level 1 reaches a reasonable conclusion but supports it with two sentences; Level 3 gives the context (flat wild catch since 1990), the advantages in detail and the impacts with ways to reduce them.	
	• Level 3 distinguishes between species types (herbivores and molluscs vs carnivores) - the key to 'to what extent'.	
	• Level 3 also considers whether fishing can be made sustainable, which the comparison requires.	
10.2	1 mark for the point below. **1** A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Energy flow:** autotrophs and heterotrophs – losses by reflection, transmission, respiration – 100 000 to 1 000 units – few food chains beyond four levels – plant food most productive – grazing animals where crops impossible – pigs on wastes • **Energetics:** intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • **Control of genetics:** asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • **GM:** Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape – advantages – risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity)	25
	• **Sustainability links:** energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Need:** population possibly 12 billion – humans raise carrying capacity – trophic level of diet • **Impacts to be reduced:** habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • **Social, economic and political influences:** consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • **Strategies:** cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • **Limits:** little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25

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